

LONG HARMONIC SUMS WITHOUT DECREASING DENOMINATOR

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ABSTRACT. Let $1 \leq a < b$ be integers, and write the sum

$$\frac{1}{a} + \frac{1}{a+1} + \cdots + \frac{1}{b}$$

as the (simplified) fraction $\frac{u_{a,b}}{v_{a,b}}$. With this definition it turns out that, for every fixed a , there exists an integer b such that $v_{a,b} < v_{a,b-1}$. Moreover, for the smallest such $b = b(a)$, the limit inferior

$$\liminf_{a \rightarrow \infty} \left(\frac{b(a) - a}{\log a} \right)$$

was recently determined. Here we show that, for almost all a , we have the lower bound

$$b(a) > a + \exp \left(\frac{1}{2} \sqrt{\log a \log \log a} \right).$$

1. INTRODUCTION

For given integers $1 \leq a < b$, define $u_{a,b}$ and $v_{a,b}$ as the unique coprime positive integers with

$$\frac{u_{a,b}}{v_{a,b}} = \sum_{i=a}^b \frac{1}{i}.$$

With $b(a)$ the smallest integer $b > a$ such that $v_{a,b} < v_{a,b-1}$, in [4] it was shown that there exists a constant $L \in [0.54, 0.55]$ such that for all $\varepsilon > 0$ there are infinitely many a with $b(a) < a + (L + \varepsilon) \log a$. Conversely, [3, Lemma 31] gives the complementary inequality $b(a) > a + (L - \varepsilon) \log a$ for all large enough a . Hence, by combining these results we find

$$\liminf_{a \rightarrow \infty} \left(\frac{b(a) - a}{\log a} \right) = L.$$

As for upper bounds, [3, Theorem 2] gives $b(a) \leq 4.374(a - 1)$ for all $a \geq 6$. This result (and, in particular, the existence of $b(a)$ in the first place) answered a question of Erdős and Graham [5], and resolved Erdős Problem #290 on Bloom's website [1]. However, in contrast to the aforementioned lower bound results, we do not believe that the bound $b(a) \leq 4.374(a - 1)$ is close to optimal. In fact, we conjecture that we actually have $b(a) < a + a^\varepsilon$ for all $\varepsilon > 0$ and all large enough a . In the other direction, one cannot lower the a^ε term all the way down to $(\log a)^C$ for any fixed constant C . Indeed, in this short note we prove the following.

Theorem 1. *For almost all a we have*

$$b(a) > a + \exp \left(\frac{1}{2} \sqrt{\log a \log \log a} \right).$$

2. MAIN RESULT AND PROOF

In order to prove Theorem 1, we need one preliminary result. To be able to state it, we define L_n as the least common multiple of the first n positive integers.

Lemma 2. *For all positive integers $a < b$ we have*

$$\frac{v_{a,b}}{v_{a,b-1}} \geq \frac{b}{\gcd(b, L_{b-a})^2}.$$

In particular, if $v_{a,b} < v_{a,b-1}$, then $\gcd(b, L_{b-a}) > \sqrt{b}$.

Proof. Let $g := \gcd(b, v_{a,b-1})$ and define $b' := \frac{b}{g}$ and $v' := \frac{v_{a,b-1}}{g}$. We then have

$$\frac{u_{a,b}}{v_{a,b}} = \frac{u_{a,b-1}}{v_{a,b-1}} + \frac{1}{b} = \frac{u_{a,b-1}b' + v'}{gb'v'}.$$

This implies $v_{a,b} \geq b'v' = \frac{bv_{a,b-1}}{g^2}$, as $\gcd(u_{a,b-1}b' + v', b'v') = 1$. It therefore suffices to show $g \leq \gcd(b, L_{b-a})$, and we will do this one prime at a time. So let q be any prime power divisor of g . Since $v_{a,b-1}$ divides $\text{lcm}(a, a+1, \dots, b-1)$, q divides some integer in the interval $[a, b-1]$. But since q also divides b , this is only possible when $q \leq b-a$. And this happens precisely when $q \mid L_{b-a}$. $\square$

We are now ready to prove our main result.

Proof of Theorem 1. Let x be sufficiently large and let n be a positive integer with

$$n \leq N := \exp\left(\frac{1}{2}\sqrt{\log 2x \log \log 2x}\right).$$

We then aim to count the number of integers a with

$$x \leq a < 2x, \quad b := a + n, \quad \text{and} \quad v_{a,b} < v_{a,b-1}.$$

For such a pair we have $d := \gcd(b, L_n) > \sqrt{b} > \sqrt{x}$ by Lemma 2. On the other hand,

$$d \leq b = a + n < 2x + x = 3x,$$

if x is large enough. Moreover, since $d \mid b = a + n$, we must have $a \equiv -n \pmod{d}$. By summing over all possible values of d , we therefore see

$$\begin{aligned} \#\{a \in [x, 2x) : v_{a,a+n} < v_{a,a+n-1}\} &\leq \sum_{\substack{d \mid L_n \\ \sqrt{x} < d < 3x}} \#\{a \in [x, 2x) : a \equiv -n \pmod{d}\} \\ &\ll x \sum_{\substack{d \mid L_n \\ d > \sqrt{x}}} \frac{1}{d}. \end{aligned}$$

In order to deal with this latter sum we recall the elementary inequality

$$-\log(1-t) \leq \frac{t}{1-t} \quad \forall t \in (0, 1),$$

which implies $-\log(1-t) < 3t$ whenever $t < \frac{2}{3}$. In particular, for any $\delta \in (0, \frac{1}{3})$, the above sum can be upper bounded using Rankin's trick by

$$\begin{aligned}
\sum_{\substack{d|L_n \\ d > \sqrt{x}}} \frac{1}{d} &< x^{-\delta/2} \sum_{d|L_n} \frac{1}{d^{1-\delta}} \\
&= x^{-\delta/2} \prod_{p^k \| L_n} \left(1 + \frac{1}{p^{1-\delta}} + \cdots + \frac{1}{p^{k(1-\delta)}} \right) \\
&< x^{-\delta/2} \prod_{p \leq n} \left(1 - \frac{1}{p^{1-\delta}} \right)^{-1} \\
&= x^{-\delta/2} \exp \left[\sum_{p \leq n} -\log \left(1 - \frac{1}{p^{1-\delta}} \right) \right] \\
&< x^{-\delta/2} \exp \left[3 \sum_{p \leq n} \frac{1}{p^{1-\delta}} \right].
\end{aligned}$$

By partial summation and (a weak version of) the prime number theorem, we have

$$\begin{aligned}
\sum_{p \leq n} \frac{1}{p^{1-\delta}} &\leq \sum_{p \leq N} \frac{1}{p^{1-\delta}} \\
&= \frac{\pi(N)}{N^{1-\delta}} + (1-\delta) \int_2^N \frac{\pi(t)}{t^{2-\delta}} dt \\
&\ll \frac{N^\delta}{\log N} + \int_2^N \frac{t^{-1+\delta}}{\log t} dt \\
&= \frac{N^\delta}{\log N} + \int_{\delta \log 2}^{\delta \log N} \frac{e^t}{t} dt.
\end{aligned}$$

Splitting this integral at $t = 1$ gives

$$\begin{aligned}
\int_{\delta \log 2}^{\delta \log N} \frac{e^t}{t} dt &= \int_{\delta \log 2}^1 \frac{e^t}{t} dt + \int_1^{\delta \log N} \frac{e^t}{t} dt \\
&\ll \log \left(\frac{1}{\delta} \right) + \frac{N^\delta}{\delta \log N}.
\end{aligned}$$

Combining this with our prior estimates and summing over all $n \leq N$, we obtain that the number of $a \in [x, 2x)$ with

$$b(a) \leq a + \exp \left(\frac{1}{2} \sqrt{\log a \log \log a} \right) < a + N$$

is at most

$$x^{1-\delta/2} N \exp \left[O \left(\log \left(\frac{1}{\delta} \right) + \frac{N^\delta}{\delta \log N} \right) \right].$$

We now choose

$$\delta := \sqrt{\frac{\log \log 2x}{\log 2x}} \left(1 + \frac{1}{\log \log 2x} \right),$$

from which one can calculate the following estimates:

$$\begin{aligned}
\log\left(\frac{1}{\delta}\right) &< \log \log 2x, \\
N^\delta &= \sqrt{e \log 2x}, \\
\delta \log N &= \frac{1}{2}(\log \log 2x + 1), \\
x^{1-\delta/2} &< (2x)^{1-\delta/2} \\
&= 2x \exp\left[-\frac{1}{2}\sqrt{\log 2x \log \log 2x} \left(1 + \frac{1}{\log \log 2x}\right)\right] \\
&= 2x N^{-1} \exp\left(-\frac{1}{2}\sqrt{\frac{\log 2x}{\log \log 2x}}\right).
\end{aligned}$$

Hence, we finally find

$$\begin{aligned}
&\#\left\{a \in [x, 2x) : b(a) \leq a + \exp\left(\frac{1}{2}\sqrt{\log a \log \log a}\right)\right\} \\
&\leq x^{1-\delta/2} N \exp\left[O\left(\log\left(\frac{1}{\delta}\right) + \frac{N^\delta}{\delta \log N}\right)\right] \\
&\leq 2x \exp\left(O\left(\log \log 2x + \frac{\sqrt{\log 2x}}{\log \log 2x}\right) - \frac{1}{2}\sqrt{\frac{\log 2x}{\log \log 2x}}\right) \\
&= o(x),
\end{aligned}$$

from which the proof is finished by dyadic decomposition. $\square$

3. DECLARATION OF AI USAGE

Although this note is human-written, credit for originally coming up with the proof of our main result goes to ChatGPT 5.6-Sol Pro. ChatGPT's write-up is still available at [2], for transparency's sake.

REFERENCES

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